

Batuhan Ayci, MD

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EDUCATION *Medical Doctor* **GPA:3.66/4.00 2018-2024**
[Istanbul University-Cerrahpasa, Cerrahpasa School of Medicine](#), Istanbul, Turkey
• Awarded an MD with high honor in September 2024.

RESEARCH EXPERIENCE *Postdoctoral Fellow* **December 2024-Present**
[Icahn School of Medicine at Mount Sinai, Department of Neurology, Beck's Lab](#), New York, New York, USA
Supervisor(s): [Erin Savner Beck, MD](#)
• Utilized 3T and 7T MRI to investigate advanced imaging markers in early multiple sclerosis, including cortical lesions, paramagnetic rim lesions, volumetrics, and quantitative MRI metrics.
• Studied advanced imaging biomarkers and natural history in radiologically isolated syndrome (RIS) to have a better understanding of the disease.
• Applied deep learning-based reconstruction techniques to enhance cortical lesion detection sensitivity at 7T MRI.
• Collaborated on the development of FLAMeS, a robust deep learning model for automated MS lesion segmentation.

Research Assistant **December 2022-November 2024**
[Istanbul University-Cerrahpasa, Cerrahpasa School of Medicine](#), Department of Neurology, Istanbul, Turkey
Supervisor(s): [Ugur Uygunoglu, MD](#); [Melih Tutuncu, MD](#); [Aksel Siva, MD, FEAN⁵](#); [Bengi Gul Alparslan Turk, MD](#)
• Authored a book chapter pertaining the subject of Neurobehcet syndrome.
• Analysed demyelinating autoantibodies and examined treatment modalities in patients with Neurobehcet syndrome.
• Compared the effects of discontinuation of fingolimod in patients with multiple sclerosis.
• Involved in a study in patients with familial multiple sclerosis.
• Participated in project writing for neuromodulation in patients with multiple sclerosis.
• Implemented [REDCap](#) to clinical practice in patients with headache and multiple sclerosis and related disorders outpatient clinics.

Research Assistant **August 2021-November 2024**
[Istanbul University-Cerrahpasa, Cerrahpasa School of Medicine](#), Department of Histology and Embryology, Istanbul, Turkey
Supervisor(s): [Gozde Erkanli Senturk, PhD](#)
• Conducted animal experiments using mice with follow-up procedures, injections, animal terminations, and collection of tissues and embryos.
• Performed immunohistochemistry, immunofluorescence, and ELISA with the consequent analysis utilizing Olympus X61 microscope and ImageJ.
• Developed and involved in neuroscience projects supported by TUBITAK¹ and BAP²
– *Project Manager* May 2022-May 2024
TUBITAK 2209-A Research Project Support Program for Undergraduate Students
– *Scholar* February 2024-July 2024
TUBITAK STAR-Intern Researcher Scholarship Program

¹The Scientific and Technological Research Council of Turkey

²University's Scientific Research Projects Coordination Unit

- *Scholar* April 2022-April 2023
TUBITAK Short Term R&D Funding Program 1002 Project
- *Researcher* July 2022-September 2023
BAP Undergraduate Student Research Project

Research Internship

December 2022-November 2024

[Istanbul University-Cerrahpasa, Cerrahpasa School of Medicine](#), Department of Internal Medicine, Division of Rheumatology, Istanbul, Turkey

Supervisor(s): [Serdal Ugurlu, MD](#)

- Designed a clinical study on the neurological aspects of familial Mediterranean fever.
- Covered research in the relationship between the familial Mediterranean fever and multiple sclerosis.
- Studied on cardiovascular comorbidities and clinical biomarkers in patients with familial Mediterranean fever. Presented in ISSAID³ Annual Meeting 2023 and EULAR⁴ 2023.
- Reviewed neurological involvement of familial Mediterranean fever.
- Commented on a study focusing on stroke in familial Mediterranean fever.

Summer Internship

June 2023-August 2023

[University of Basel, School of Medicine, Department of Biomedical Engineering](#), Basel, Switzerland

Supervisor(s): [Cristina Granziera, MD, PhD](#); [Alessandro Cagol, MD](#)

- Studied MRI images in patients with multiple sclerosis.
- Focused on paramagnetic rim lesions and cortical lesions with variable conventional and quantitative MRI utilizing ITK-SNAP software.

**CLINICAL
EXPERIENCE**

Volunteer Internship

December 2022-September 2024

[Istanbul University-Cerrahpasa, Cerrahpasa School of Medicine](#), Department of Neurology, Istanbul, Turkey

Supervisor(s): [Ugur Uygunoglu, MD](#); [Melih Tutuncu, MD](#); [Aksel Siva, MD, FEAN⁵](#); [Sabahattin Saip, MD](#); [Bengi Gul Alparslan Turk, MD](#)

- Participated in the provision of the patient care in a clinic specializing multiple sclerosis and related disorders.
- Engaged in educational and clinical multiple sclerosis meetings, discussing the management of patients.
- Exhibited communication skills through effective engagement with patients, their families, and fellow healthcare professionals, thereby facilitating comprehensive delivery of care.

Visiting Medical Student Clerkship

May 2024-May 2024

[Mayo Clinic, College of Medicine and Science, Department of Neurology](#), Rochester, Minnesota, USA

Supervisor(s): [Eoin Flanagan, MB, BCh, FAAN⁶](#); [Zafer Keser, MD](#); [Michel Toledano, MD](#); [Nicholas Niall Madigan, MB, BCh, PhD](#); [Eric James Sorenson, MD, FAAN⁶](#); [Teerin Liewluck, MD, FAAN⁶](#); [Margherita Milone, MD, PhD, FAAN⁶](#)

- Engaged in delivering patient care across both inpatient and outpatient settings, conducting detailed patient histories and physical examinations.
- Developed proficiency in documenting patient encounters, writing detailed clinical notes, and formulating evidence-based treatment plans.
- Took part in multidisciplinary team meetings, discussing patient cases and contributing to the development of individualized care plans.
- Demonstrated strong communication skills by effectively interacting with patients, families, and healthcare team members to ensure comprehensive care delivery.

³The International Society of Auto-Inflammatory Diseases

⁴European Alliance of Associations for Rheumatology

⁵Fellow of the European Academy of Neurology

⁶Fellow of the American Academy of Neurology

- Participated in educational sessions and clinical rounds, expanding my knowledge of the latest advancements and research in neurology.

Observership

March 2024-May 2024

National Institutes of Health, National Institute of Neurological Disorders and Stroke, Neuroimmunology Clinic, Bethesda, Maryland, USA

Supervisor(s): Daniel Salo Reich, MD, PhD; Maria Ines Gaitan, MD

- Engaged in delivering patient care within a clinic focused on multiple sclerosis, progressive multifocal leukoencephalopathy, and HTLV-1-associated myelopathy/tropical spastic paraparesis.
- Participated in gatherings centered around neuroimmunology as well as weekly neurological meetings, engaging in discussions pertaining to patient management strategies.
- Gained experience in presenting and documenting patient encounters, crafting clinical notes.

PUBLICATIONS

6. Karup S, Baspinar SB, **Ayci B**, Azman Tunc FN, Akyel A, Guler Y, Ugurlu S. [Familial Mediterranean Fever and Cardiovascular Risk](#). *Mediterr J Rheumatol*. 2026;37(2):415-420. Published on June 1st, 2026.
5. **Ayci B**, Dereskewicz E, dos Santos Silva J, Galasso J, Rust P, La Rosa F, Liu J, Reich DS, Sumowski JF, Beck ES. [Cortical Lesions Form Predominantly in Early Multiple Sclerosis](#). Preprint. medRxiv. 2026;2026.04.30.26352141. Published on May 1st, 2026.
4. Cagol A, Benkert P, Schaedelin S, Ocampo-Pineda M, Montobbio N, Lu PJ, **Ayci B**, Wenger A, Shukur AA, Kaim K, Melie-Garcia L, Weigel M, Signori A, Calabrese P, Kappos L, Sormani MP, Kuhle J, Granziera C. [Assessing the Relative Importance of Imaging and Serum Biomarkers in Capturing Disability, Cognitive Impairment, and Clinical Progression in Multiple Sclerosis](#). *Adv Sci (Weinh)*. 13(10):e12946. 10.1002/adv.202512946. Published on January 12th, 2025.
3. Dereskewicz E, La Rosa F, Dos Santos Silva J, Sizer E, Kohli A, Wynen M, Mullins WA, Maggi P, Levy S, Onyemeh K, **Ayci B**, Solomon AJ, Assländer J, Al-Louzi O, Reich DS, Sumowski J, Beck ES. [A Novel Convolutional Neural Network for Automated Multiple Sclerosis Brain Lesion Segmentation](#). *J Neuroimaging*. 2025;35(5):e70085. 10.1111/jon.70085. Published on September 12th, 2025.
2. **Ayci B**, Gulec B, Uygunglu U, Siva, A. [Neurobehcet Syndrome \[in Turkish\]](#). In: Yucesan C, Tuncer A, Uzunkopru C. *Neuroimmunology*. 1st ed. Ankara: Türk Nöroloji Derneği Yayınları; 2023:261-276.
1. Silici S, Özer S, Okan A, **Ayci B**, Soylu M, Doğanyigit Z. [Histological, Hematological and Biochemical Effects of Bee Bread on DSS-Induced Colitis Model in Rats](#). *EC Pharmacology and Toxicology*. 2023;10(4):43-52.

PRESENTATIONS

22. Cheshire M, Dereskewicz E, **Ayci B**, Dos Santos Silva J, Feng Q, Galasso J, La Rosa F, Sumowski JF, Beck ES. [Regional Cortical Demyelination Predicts Word-Finding Difficulty in Early Multiple Sclerosis](#). Poster presented at: 2026 American Academy of Neurology Annual Meeting, April 2026, Chicago, Illinois, USA.
21. **Ayci B**, Dereskewicz E, Silva JD, Galasso J, La Rosa F, Liu J, Reich DS, Sumowski JF, Beck ES. [P272 Cortical Lesions Constitute a Substantial Portion of Overall Lesion Burden and Relate to Cognitive Performance in Early MS](#). Poster presented at: Americas Committee for Treatment Research in Multiple Sclerosis (ACTRIMS) Forum 2026, February 2026, San Diego, California, USA.
20. Dereskewicz E, **Ayci B**, Yu T, La Rosa F, Piredda GF, Nickel MD, Dos Santos Silva J, Liebig P, Thiran J, Kober T, Hilbert T, Beck ES. [P245 Deep Learning-Based Reconstruction of 7T MP2RAGE MRI for Multiple Sclerosis Cortical Lesion Visualization](#). Poster presented at: Americas Committee for Treatment Research in Multiple Sclerosis (ACTRIMS) Forum 2026, February 2026, San Diego, California, USA.
19. Cheshire M, Dereskewicz EM, **Ayci B**, Dos Santos Silva J, Feng Q, Galasso F, La Rosa F, Sumowski JF, Beck ES. [P263 Linking Local Cortical Demyelination to Language Impairment in Multiple Sclerosis](#). Poster presented at: Americas Committee for Treatment

Research in Multiple Sclerosis (ACTRIMS) Forum 2026, February 2026, San Diego, California, USA.

18. Aoun R, Nee A, Sizer E, **Ayci B**, Dereskewicz E, Onyemeh K, La Rosa F, Sati P, Sumowski JF, Beck ES. [P302 Paramagnetic rim lesions are associated with decreased cognitive speed in a large, early MS cohort](#). Poster presented at: Americas Committee for Treatment Research in Multiple Sclerosis (ACTRIMS) Forum 2026, February 2026, San Diego, California, USA.
17. Cagol A, Ocampo-Pineda M, Benkert P, **Ayci B**, Lu PJ, Schadelin S, Montobbio N, Melie-Garcia L, Weigel M, Galbusera R, Ruberte E, Chen X, Kappos L, Kuhle J, Sormani MP, Granziera C. [Unraveling the Pathological and Clinical Heterogeneity of Paramagnetic Rim Lesions in Multiple Sclerosis](#). *Mult Scler.* 2025;31(3 Suppl):588-589.
16. Spagnolo F, Bhardwaj A, Gordaliza PM, Lu PJ, Ocampo-Pineda M, Bach Cuadra M, Chen X, **Ayci B**, Cagol A, Andrearczyk V, Depeursinge A, Granziera C. [Deep-PRL: A Deep Learning Network for the Identification of Paramagnetic Rim Lesions in Multiple Sclerosis](#). Oral presented at: International Society for Magnetic Resonance in Medicine (ISMRM) and International Society for MR Radiographers Technologists (ISMRT) Annual Meeting and Exhibition; May 2025; Honolulu, HI, USA.
15. **Ayci B**, Gorkey OD, Everest E, Tutuncu M, Saip S, Siva A, Uygunoglu U. [Adherence to Dimethyl Fumarate in Treatment-Naive and Therapy-Switched Individuals with Multiple Sclerosis: A Comparative Study](#). *Neurology.* 2025;104(7 Suppl 1):4665.
14. Iris M, Tutuncu M, **Ayci B**, Tutuncu M, Uygunoglu U, Saip S, Ugurlu S, Siva A. [Comorbidity of Familial Mediterranean Fever with Multiple Sclerosis](#). *Neurology.* 2025;104(7 Suppl 1):5007.
13. Karup S, Baspinar SN, **Ayci B**, Azman FN, Akyel A, Guler Y, Ugurlu S. [PO130 Cardiovascular Comorbidities in Patients with Familial Mediterranean Fever](#). Poster presented at: International Society of Systemic Auto-inflammatory Diseases Annual Meeting 2025; April 2025; Paris, France.
12. **Ayci B**, Gorkey OD, Everest E, Tutuncu M, Saip S, Siva A, Uygunoglu U. [Dimethyl Fumarate Compliance in the Treatment of Multiple Sclerosis: Cerrahpasa Experience \[in Turkish\]](#). Oral presented at: 60th National Neurology Congress. December 2024; Antalya, Turkey.
11. Ugurlu S, **Ayci B**, Karup S, Aktas E, Kilinc OC, Ates MB, Azman FN, Yenidogan I, Balci AS, Ozdogan H. [AB0200 Redness Over a Joint as a Clinical Marker in Patients with Familial Mediterranean Fever](#). *Ann Rheum Dis.* 2024, 83(Suppl 1):1337.
10. Cagol A, Ocampo-Pineda M, **Ayci B**, Benkert P, Lu P, Weigel M, Melie-Garcia L, Chen X, Lutti A, Nguyen TD, Wang Y, Lee J, Kuhle J, Kappos L, Sormani MP, Granziera C. [Multiparametric Quantitative MRI Shows Enhanced Degeneration in Paramagnetic Rim Lesions and Their Perilesional Tissue in Multiple Sclerosis](#). Oral presented at: 2024 International Society for Magnetic Resonance in Medicine (ISMRM) and International Society for MR Radiographers Technologists (ISMRT) Annual Meeting and Exhibition; May 2024; Singapore, Australia.
9. Karup S, Baspinar SN, **Ayci B**, Azman FN, Akyel A, Guler Y, Ugurlu S. Cardiovascular Comorbidities in Patients with Familial Mediterranean Fever Over 45 Years of Age. Oral presented at: 2nd Familial Mediterranean Fever Meeting; May 2024; Istanbul, Turkey.
8. Gur E, **Ayci B**, Karup S, Guler A, Akyel A, Doga G, Gul Turk B, Asan F, Uygunoglu U, Ugurlu S. [SS-14 Investigation of headache characteristics in patients with Familial Mediterranean Fever \[in Turkish\]](#). Oral presented at: 1. National Headache Congress. March 2024; Istanbul, Turkey.
7. Cagol A, Ocampo-Pineda M, Benkert P, **Ayci B**, Lu P, Melie-Garcia L, Schadelin S, Weigel M, Leber S, Galbusera R, Ruberte E, Chen X, Barakovic M, Yaldizli O, Oechtering J, Lorscheider J, D'souza M, Fischer-Barnicol B, Mueller S, Achtnichts L, Vehoff J, Disanto G, Findling O, Chan A, Pot C, Bridel C, Zecca C, Derfuss T, Lieb1 J, Remonda L, Vargas MI, Wagner F, Du Pasquier R, Lalive P, Pravata E, Weber J, Cattin PC, Hoepner R, Roth P, Gobbi C, Leppert D, Lutti A, Nguyen T, Wang Y, Lee J, Kuhle J, Kappos L, Sormani M, Granziera C. [V243. Paramagnetic Rim Lesions Associate With Enhanced Microstructural and Diffuse Neurodegenerative Damage in People With Multiple Sclerosis](#). Poster presented at: Americas Committee For Treatment and Research in

Multiple Sclerosis (ACTRIMS) 2024. February-March 2024; West Palm Beach, Florida, USA.

6. Taskin N, Sahin H, **Ayci B**, Erkanli Senturk G. High dose folic acid may alter the filtration barrier of glomeruli in late pregnancy period. Oral presented at: 2024 14th International Conference on Bioscience, Biochemistry and Bioinformatics (ICBBB 2024); January 2024; Kyoto, Japan.
5. Taskin N, Sahin B, **Ayci B**, Erkanli Senturk G. [16-Histological Assessment of Renal Effects in Mice Exposed to High Dose Folic Acid During Different Pregnancy Days](#). Oral presented at: 26th National Electron Microscopy Congress; September 2023; Eskisehir, Turkey.
4. Karup S, **Ayci B**, Baspinar SN, Azman FN, Ugurlu S. [POS1571 Cardiovascular Comorbidities in Patients with Familial Mediterranean Fever Over 50 Years of Age](#). Ann Rheum Dis. 2023;82(Suppl 1):1161.
3. Baspinar SN, Karup S, **Ayci B**, Guler Y, Akyel A, Can G, Ugurlu S. [P145 Cardiovascular Comorbidities in Patients with Familial Mediterranean Fever Over 45 Years of Age](#). Poster presented at: The International Society of Systemic Auto-Inflammatory Diseases Annual Meeting 2023; May 2023; Toronto, Canada.
2. Cetin KI, **Ayci B**, Kose SN, Ser MH, Gul Turk B, Poyraz BC, Uygunoglu U. [Investigation of Personality Characteristics of Patients Diagnosed with Medication Overuse Headache](#). Poster presented at: 8th International Medical Student Research Congress; April 2023; Istanbul, Turkey.
1. Iris M, **Ayci B**, Cetin KI, Kose SN, Ser MH, Gul Turk B, Poyraz BC, Uygunoglu U. [SS-19 Investigation of Personality Characteristics of Patients Diagnosed with Medication Overuse Headache \[in Turkish\]](#). Oral presented at: 19th Headache Winter School; April 2023, Istanbul, Turkey.

EXTRA-CURRICULARS & VOLUNTEER EXPERIENCES	American Academy of Neurology	
	• Active Member	December 2022-Present
	Turkish Neurological Society	
	• Active Member	January 2023-Present
	European Academy of Neurology	
	• Active Member	January 2023-December 2024
	Society of Turkish American Physicians	
	• Active Member	April 2024-Present
	Cerrahpasa School of Medicine Social Support Volunteers for Earthquake Victims	
	• Active Volunteer	February 2023-September 2023
	Aid Organization For Turkey-Syria Earthquake	
	• Active Volunteer	February 2023-March 2023
	Scientific Project Community	
	• Member	January 2021-June 2021
	Cerrahpasa School of Medicine Photography Club	
	• President	September 2020-October 2021
CERTIFICATES & LICENSURES	• Member	September 2018-September 2024
	Pass in USMLE ⁷ Step 1	October 2022
	Yeditepe University Certificate of Animal Use in Experimental Research	October 2022
	Basic Life Support Course from American Heart Association	October 2023
	Individual License from Turkish Cycling Federation	March 2023-December 2023

⁷The United States Medical Licensing Examination

AWARDS & SCHOLARSHIPS	Summit Best Poster Award in ACTRIMS ⁸	February 2026
	European Academy of Neurology Student Teaser Fellowship 2023	May 2023
	5th place in Presidency Digital Transformation Office & Turksat Ideathon	December 2019
	735th among more than 2,5 million in the national university entrance exam	July 2018
LANGUAGE & OTHER SKILLS	Turkish (Native)	
	English (Full professional proficiency)	
	107/120 (R:27, L:26, S:27, W:27) in TOEFL⁹	October 2023
	C1 level English proficiency certificate	August 2019
	– Obtained after attending an English course for 2 months at the Centre of English Studies in Worthing, the UK.	
	87.5/100 in YDS¹⁰	July 2019
PERSONAL INTERESTS & HOBBIES	Cycling, running, and trekking.	
	Interested in history, philosophy, and theatre.	
	Keen on photography and beer tasting.	
	Used to play guitar.	

⁸Americas Committee for Treatment Research in Multiple Sclerosis

⁹Test of English as a Foreign Language

¹⁰The Foreign Language Knowledge Level Determination Exam, organised by the state in Turkey